

File 14524: Source:  $\sigma_{sky}$  (Astro)

\* CCD:  $\frac{S}{N} \sim \frac{t N_s}{\sqrt{N_s t + L}} \leftarrow N_s$

$\uparrow$  (gave when only when  $n_{pix} = 1$ )  
 $\nwarrow$

$n_{pix} \approx 1$  ( )

$$\sigma = \sqrt{N}$$

$$\frac{S}{N} = \frac{N_s t}{\sqrt{N_s t + n_{pix}(n_s t + n_d t + n_r^2)}}$$

Q 12: linear expansion coeff.

$$\frac{\Delta L}{L_0} = \alpha \Delta T$$

$\Delta L$ : change in length

$\Delta T$ : change in temp

$L_0$ : original length

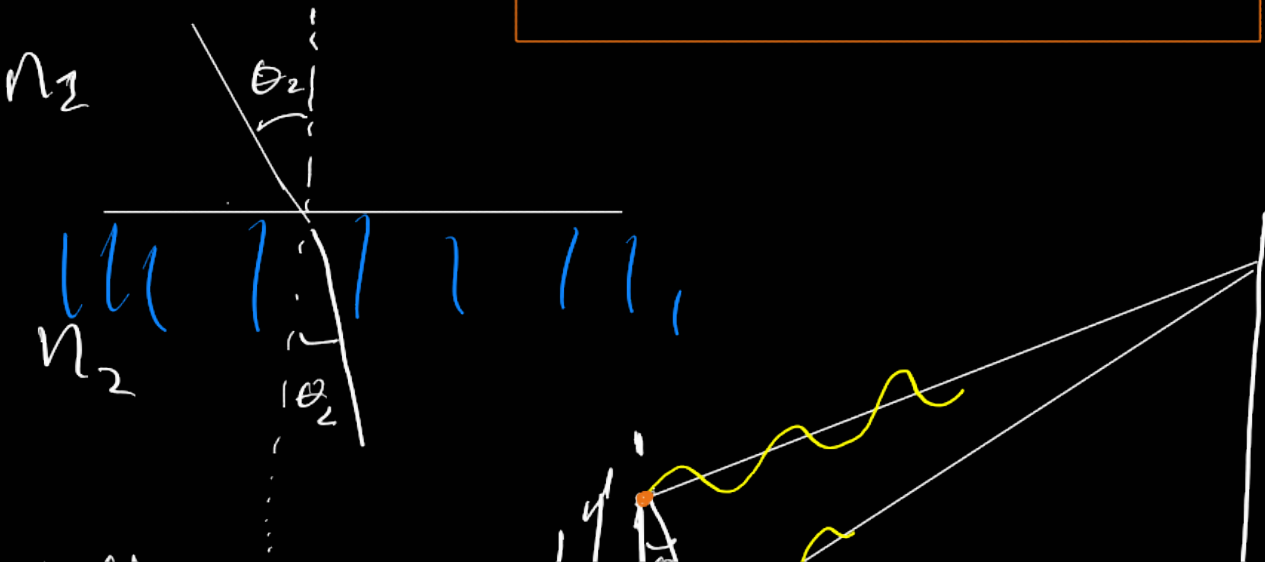
Vol:  $\frac{\Delta V}{V_0} = \beta \Delta T$

(thermo)

File 13524: (light/Diffraction/optics)

Snell's law:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$



diffraction:



$$\sin(\theta) = \frac{l}{d}$$

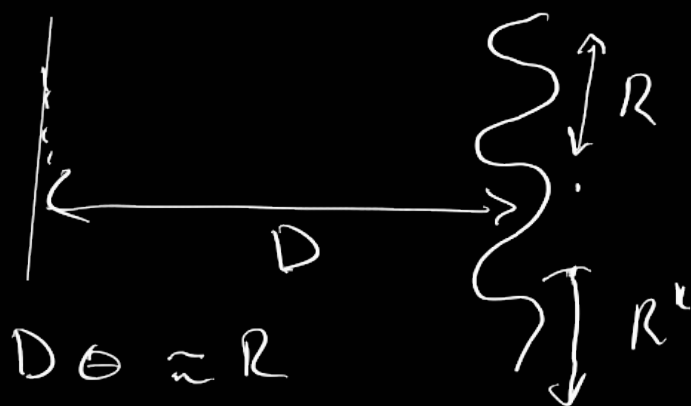
•  $l = d \sin(\theta)$

two cases:

(i) Constructive:  $l = n\lambda$

$$d \sin(\theta) = n\lambda, \quad n = 0, 1, 2, 3, \dots$$

(ii) destructive:  $d \sin(\theta) = (n + \frac{1}{2})\lambda$



File 10524:

② (Dim analysis)  $\chi(r, \frac{p}{e}, n)$

③ (Mechanics) Energy conservation + Moments  
→ Work: from friction (from not slipping)  $K.E. = \frac{1}{2} \omega x^2$

④ (Semi-classical/light)



$\Delta\lambda$ : bandwidth/wavelength of the wave length/variation

$$* E = hf = \frac{hc}{\lambda}$$

$\Delta E, \Delta \lambda$  : taking a derivative

$$|\Delta E| = \frac{hc}{\lambda^2} \Delta \lambda$$

Uncertainty principle:

$$\Delta E \Delta t \approx \frac{\hbar}{2}$$

$$\Delta x \Delta p \approx \frac{\hbar}{2}$$

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File 9524: (Quantum)

2022 Q9: 3D infinite square well

$$E_N = \frac{\hbar^2 \pi^2}{2m\omega^2} N, \quad N = n_x^2 + n_y^2 + n_z^2$$

1D Square well:

$$E_n = \frac{\hbar^2 \pi^2}{2m\omega^2} n^2$$

$$n_x, n_y, n_z = 0, 1, 2, \dots$$

degeneracies: The # of states which have the same energy levels

ex:  $N=1 = n_x^2 + n_y^2 + n_z^2$

3 cases:

- (i)  $n_x = 1$ , rest 0
- (ii)  $n_y = 1$ , rest 0
- (iii)  $n_z = 1$ , rest 0

$E_1$  : 3 cases

Q 10: (Probability:)

$(|\psi|^2 dx)$ : prob that  $\psi$  is found between  $x$  and  $(x + dx)$

$\int_a^b |\psi|^2 dx$ : prob that  $\psi$  is between  $a$  and  $b$

$\int_{-\infty}^{\infty} |\psi|^2 dx$ : (total prob) = 1

Spherical coordinates:

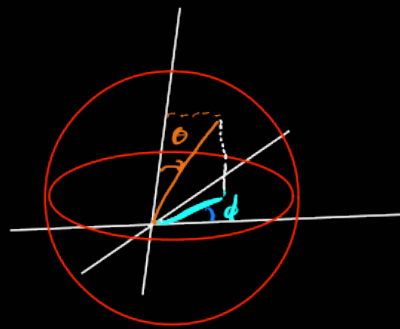
$$dx dy dz = r^2 \sin(\theta) dr d\theta d\phi$$

$$\int_{-\infty}^{\infty} dx dy dz = \int_0^{\pi} \sin(\theta) d\theta \int_0^{2\pi} d\phi \int_0^{\infty} r^2 dr$$

$$x = r \cos(\phi) \sin(\theta)$$

$$y = r \sin(\theta) \sin(\phi)$$

$$z = r \cos(\theta)$$



Q3:  $\frac{\partial^2 f}{\partial t^2} = -\omega^2 f$

(Quantum)

Simple Harmonic Motion

Soln:  $f = C e^{\pm i\omega t}$

↑  
Const.

$\frac{\partial^2 f}{\partial t^2} = +\omega^2 f$  | Soln:  $f = C e^{\pm \omega t}$

8524: trig identities: (see images)

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

# (E S M) Maxwells Equations + Lorentz force

Lorentz force:  $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$  (test part)  
 $\vec{v}$ : velocity  
 $\vec{B}$ : B-field  
 $\vec{E}$ : E-field

Maxwell's:

Differential form

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \leftarrow \text{charge density}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$\vec{J}$ :  
(current density)

Integral form

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

$dA$ : Area

$$\oint \vec{E} \cdot d\vec{l} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{A}$$

$\downarrow$   
length

$$\oint \vec{B} \cdot d\vec{A} = 0$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} + \mu_0 \epsilon_0 \int \frac{\partial \vec{E}}{\partial t} \cdot d\vec{A}$$

$\int \vec{E} \cdot d\vec{S} = -\frac{d\Phi}{dt}$  Potential  
 Ampere's law

## (Astro) Keplers law

$$P^2 = \frac{4\pi^2}{GM} a^3$$

$\downarrow$                        $\uparrow$                        $\downarrow$   
 Period                      "main body"                      semi-major axis

Luminosity:

$$(Flux) = \frac{L}{4\pi P^2}$$

$$L = \frac{\Delta E}{\Delta t}$$

Blackbody:  $L = \sigma A T^4$   $\leftarrow$  temp.

$\uparrow$   
 Stefan-Boltzmann

$\nwarrow$  Surface Area

Wien's law:  $\lambda_{\text{peak}} T = 2.898 \times 10^{-3} \text{ m K}$

$\uparrow$   
 temp of  
 B.B.

> Only for when  $M \gg m$ . When both orbit each other

$$P^2 = \frac{4\pi}{G(M+m)} a^3$$

Telescopes: Resolution:  $R = \frac{134}{D}$

angular resolution:

$D$ : Diameter of obj.

$$\theta \approx 1.22 \frac{\lambda}{\tilde{D}}$$

$\tilde{D}$ : Diameter of telescope/lens aperture

$\lambda$ : wavelength of light

Elliptical orbits:

(Thermo) Prob. of a state having energy  $E_n$

$\uparrow$  is Boltzmann

$$P_n = \frac{e^{-E_n/k_B T}}{Z}$$

internal Energy

$$Z = \sum_n e^{-E_n/k_B T}$$

[free energy] :  $F = -k_B T \ln(Z)$

(entropy) :  $S = - \left( \frac{\partial F}{\partial T} \right)_V = \frac{U - F}{T}$

(pressure)  $P = - \left( \frac{\partial F}{\partial T} \right)_T$

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### (General Maths)

(i) Sphere :  $\begin{cases} (\text{Vol}) = \frac{4}{3} \pi R^3 \\ \downarrow \\ (\text{Surface Area}) = 4 \pi R^2 \end{cases}$

Circle :  $\begin{cases} (\text{Area}) = \pi R^2 \\ (\text{Circumference}) = 2 \pi R \end{cases}$

